ModelIt!

Student Activity Pack

Survival of the Fittest: Can You Outrun Natural Selection?

How Environmental Pressure and Genetic Diversity Shape Who Survives

Grade 08L2 | NGSS MS-LS4-4, MS-LS4-6 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What does 'survival of the fittest' mean to you?

2 Why do you think some species go extinct while others survive environmental changes?

3 What is genetic diversity, and why might it matter for a population?

4 Draw what you think happens to a population when its environment suddenly changes.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Environmental Pressure | Population Diversity | Favorable Trait Frequency | Survival Rate | Reproduction Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Stable Environment

Moderate pressure, moderate diversity

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Low Diversity Crisis

High Environmental Pressure, low Population Diversity

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High Diversity Advantage

High Environmental Pressure, high Population Diversity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Migration Rate — New individuals joining the population bring different genetic traits

Mutation Rate — Random DNA changes introduce new traits over generations

Predator Efficiency — How effectively predators target specific traits in the prey population

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can a population survive when the environment changes faster than it can adapt? Use evidence from your simulation to support your answers.

1. **Why is genetic diversity sometimes called an 'insurance policy' for a species?**

2. **How does natural selection differ from random chance? Use evidence from your model.**

3. **Why might a population that survived one environmental change still be vulnerable to the next?**

4. **How does the concept of favorable trait frequency help explain why some endangered species recover while others don't?**

5. **What would happen if humans artificially reduced the genetic diversity of a crop species?**

8 STEM Challenge: Design a Species Conservation Strategy

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YOUR ENGINEERING MISSION

Design an evidence-based conservation plan for an endangered species facing rapid environmental change, using your model data to justify every recommendation.

THE SCENARIO

A population of desert tortoises is declining as temperatures rise and drought intensifies. Wildlife biologists need your team to propose a conservation strategy that maximizes the species' chances of long-term survival.

Engineering Design Thinking

1. WHAT SPECIFIC ACTIONS WILL INCREASE GENETIC DIVERSITY IN THE POPULATION?
2. HOW WILL YOU REDUCE ENVIRONMENTAL PRESSURE ON THE SPECIES?
3. WHAT MONITORING SYSTEM WILL TRACK TRAIT FREQUENCY CHANGES OVER TIME?
4. HOW WILL YOU MEASURE WHETHER YOUR STRATEGY IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Conservation Biologists and Wildlife Geneticists study how genetic diversity affects species survival and design strategies to protect endangered populations. They earn $55,000-$95,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In your model, a population with 80% genetic diversity survived a severe drought, while a population with 20% diversity collapsed. Which explanation is best supported by the model evidence?

- **A.** The diverse population evolved new drought-resistant traits during the drought
- **B.** The diverse population already contained individuals with drought-resistant traits that natural selection could favor
- **C.** The diverse population was physically stronger and could endure more heat
- **D.** The diverse population cooperated better to share water resources

2 Your model shows that Favorable Trait Frequency increased from 15% to 80% over 10 generations under high Environmental Pressure. Which component relationship explains this shift?

- **A.** Environmental Pressure directly created favorable traits in the population
- **B.** Survival Rate increased because the environment became less harsh over time
- **C.** Individuals with favorable traits survived and reproduced at higher rates, passing those traits to offspring
- **D.** Reproduction Rate decreased, which concentrated favorable traits in fewer individuals

3 A student claims that natural selection is random because mutations are random. Using evidence from your model, which response best addresses this claim?

- **A.** The student is correct because all aspects of evolution are random
- **B.** The student is partially correct because both mutations and selection are random processes
- **C.** The student is incorrect because while mutations are random, natural selection is directional, consistently favoring traits that match the environment
- **D.** The student is incorrect because mutations are not random either

4 Two populations face the same environmental pressure of 90%. Population A has 80% diversity and stabilizes at 70% favorable trait frequency. Population B has 20% diversity and crashes toward extinction. What does this model evidence reveal about conservation?

- **A.** Endangered species should be moved to environments with less pressure
- **B.** Maintaining genetic diversity in a population is critical because it provides the variation natural selection needs to drive adaptation
- **C.** Population size is more important than genetic diversity for survival
- **D.** Populations always adapt if given enough time, regardless of diversity

5 **In your model, Reproduction Rate feeds back into Favorable Trait Frequency, creating a cycle. What does this feedback relationship represent in biological terms?**

- **A.** Organisms that reproduce more always create offspring with new, better traits
- **B.** Surviving individuals with favorable traits reproduce and increase the proportion of favorable traits in the next generation
- **C.** Higher reproduction rate causes the environment to become less harsh
- **D.** Reproduction rate has no effect on which traits become more common